



CUET ALGEBRA

COMPLETE ANALYSIS



Quadratic Equation [06]

01. Given below are two statements : [CUET 2023]
Statement I : If the roots of the quadratic equation $x^2 - 4x - \log_3 a = 0$ are real, then the least value of a is $1/81$.
Statement II : The harmonic mean of the roots of the equation $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + (8 + 2\sqrt{5}) = 0$ is 2.
 In the light of the above statements, choose the correct answer from the options given below :
 (1) Both Statement I and Statement II are true (2) Both Statement I and Statement II are false
 (3) Statement I is true but Statement II is false (4) Statement I is false but Statement II is true
02. If every pair from among the equation $x^2 + px + qr = 0$, $x^2 + qx + rp = 0$ and $x^2 + rx + pq = 0$ has a common root, then the product of three common root is _____. [CUET 2023]
 (1) pqr (2) $2pqr$ (3) $p^2q^2r^2$ (4) p^2qr^2
03. The value of x satisfies the inequality $|x - 1| + |x - 2| \geq 4$ if [CUET 2024]
 (1) $x \in \left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{7}{2}, \infty\right)$ (2) $x \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$
 (3) $x \in \left[-\frac{1}{2}, \frac{7}{2}\right]$ (4) $x \in \left(-\frac{1}{2}, \frac{7}{2}\right)$
04. If $x^2 + \frac{1}{x^2} = 2$ then the value of $x^{256} + \frac{1}{x^{256}}$ is [CUET 2024]
 (1) 1 (2) 0 (3) -2 (4) 2
05. If the roots of the equation $x^2 + 4x + a^2 - 3a = 0$ are real then the value of a (is / are) [CUET 2024]
 (1) $a \in (-\infty, -1) \cup (4, \infty)$ (2) $a \in (-\infty, -1] \cup [4, \infty)$ (3) $a \in [-2, 4]$ (4) $a \in [-1, 4]$
06. Let $\alpha > 2$ is an integer. If there are only 10 positive integers satisfying the inequality $(x - \alpha)(x - 2\alpha)(x - \alpha^2) < 0$ then the value/s of α is [CUET 2024]
 (1) 3 and 4 (2) 3 (3) -3 (4) 4
07. If $(x - 1)$ is a factor of $2x^2 - 5x + k = 0$, then the value of k is: [CUET 2025]
 (1) 2 (2) 5 (3) 3 (4) 4
08. If $x = (2 + \sqrt{3})^{\frac{1}{3}} + (2 + \sqrt{3})^{-\frac{1}{3}}$ and $x^3 - 3x + k = 0$, then the value of k is: [CUET 2025]
 (1) -4 (2) 4 (3) $\sqrt{3}$ (4) $2\sqrt{3}$



CUET ALGEBRA

COMPLETE ANALYSIS



Progression [10]

01. The a , b , c and d are in GP and are in ascending order such that $a + d = 112$ and $b + c = 48$. If the GP is continued with a as the first term, then the sum of the first six terms is [CUET 2022]

(1) 1156 (2) 1256 (3) 1356 (4) 1456

02. Match List I with List II [CUET 2022]

List I

A. If 4th term of a G.P. is square of its second term, and its first term is 3, then common ratio is _____.

B. The first term of an AP is 5, the last term is 45 and the sum of the terms is 400. The number of term is _____

C. The sum of three numbers which are in AP is 27 and sum of their squares is 293. Then the common difference is _____.

D. The fourth and 54th terms of an AP are, respectively 64 and -61. The common difference is _____.

Choose the correct answer from the options given below:

(1) A – IV; B – III; C – I; D – II

(2) A – III; B – II; C – I; D – IV

(3) A – II; B – III; C – I; D – IV

(4) A – II; B – I; C – III; D – IV

03. The arithmetic means of two observations is 125 and their geometric means is 60. Find the harmonic mean of the two observations. [CUET 2022]

(1) 4.17 (2) 8.34 (3) 28.8 (4) 57.6

04. The terms 1 , $\log_y(x)$, $\log_z(y)$ and $-15 \log_x(z)$ are in AP. Based on this information answer the following questions. [CUET 2022]

The common difference of AP is:

(1) 2 (2) -2 (3) $1/2$ (4) $-1/2$

05. The terms 1 , $\log_y(x)$, $\log_z(y)$ and $-15 \log_x(z)$ are in AP. Based on this information answer the following questions. [CUET 2022]

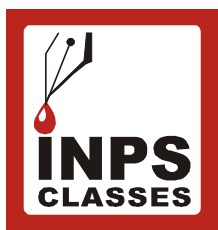
The value of xy is:

(1) 1 (2) -1 (3) z^2 (4) z^3

06. The terms 1 , $\log_y(x)$, $\log_z(y)$ and $-15 \log_x(z)$ are in AP. Based on this information answer the following questions. [CUET 2022]

yz is equal to

(1) x (2) x^2 (3) z^{-2} (4) z^{-3}



CUET ALGEBRA

COMPLETE ANALYSIS



07. Match List I with List II

[CUET 2022]

- | List I | List II |
|---|----------|
| A. In a GP, the third term is 24 and 6 th term is 192.
The common ratio is _____ | I. 26 |
| B. Let S_n denote the sum of the first n terms of an AP.
If $S_{2n} = 3S_n$, then S_{3n}/S_n equals to _____. | II. 6 |
| C. The sum of the first 3 terms of a GP is 13/12
and their product is -1. The first term is _____. | III. 4/3 |
| D. The least value of n for which the sum
$3 + 6 + 9 + \dots + 3n$ is greater than 1000 is | IV. 2 |

Choose the correct answer from the options given below:

- | | |
|------------------------------------|------------------------------------|
| (1) A – III; B – I; C – II; D – IV | (2) A – III; B – IV; C – I; D – II |
| (3) A – IV; B – II; C – III; D – I | (4) A – IV; B – III; C – II; D – I |

08. Given below are two statements : One is labeled as Assertion A and the other is labelled as Reason R.

[CUET 2023]

Assertion A : If the A.M. and G.M. between two numbers are in the ratio $m : n$, then the numbers are in the ratio $m + \sqrt{m^2 - n^2} : m - \sqrt{m^2 - n^2}$

Reason R : If each term of a G. P. be raised to the same power, the resulting sequence also forms a G.P.

In the light of the above statements, choose the correct answer from the options given below :

- | | |
|---|------------------------------|
| (1) Both A and R are true and R is the correct explanation of A | (4) A is false but R is true |
| (2) Both A and R are true but R is not the correct explanation of A | |
| (3) A is true but R is false | |

09. The H.M. of two numbers is 4 and the arithmetic mean A and geometric mean G satisfy the relation $2A + G^2 = 27$, the numbers are

[CUET 2023]

- | | | | |
|----------|----------|------------|-----------|
| (1) 6, 3 | (2) 5, 4 | (3) 5, -25 | (4) -3, 1 |
|----------|----------|------------|-----------|

10. If A_1, A_2 be two AM's and G_1, G_2 be two GM's between a and b, then $\frac{A_1 + A_2}{G_1 G_2}$ is equal to

[CUET 2023]

- | | | | |
|-----------------------|-----------------------|----------------------|-----------------------------|
| (1) $\frac{a+b}{2ab}$ | (2) $\frac{2ab}{a+b}$ | (3) $\frac{a+b}{ab}$ | (4) $\frac{a+b}{\sqrt{ab}}$ |
|-----------------------|-----------------------|----------------------|-----------------------------|

11. If a, b and c are in Geometric Progression and $a^{\frac{1}{x}} = b^{\frac{1}{y}} = c^{\frac{1}{z}}$ then, x, y, z are in _____ in

[CUET 2025]

- | | |
|---|---------------------------|
| (1) Arithmetic Progression | (2) Geometric Progression |
| (3) $\frac{2}{y} = \frac{1}{x} + \frac{1}{z}$ | (4) $x = y + z$ |



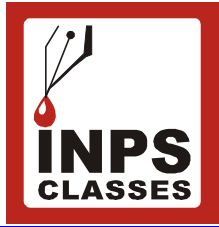
CUET ALGEBRA

COMPLETE ANALYSIS



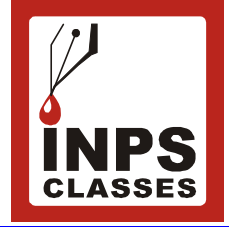
Matrices [05]

01. If $A = \begin{bmatrix} \cos B & -\sin B \\ \sin B & \cos B \end{bmatrix}$ then $A + A^T = I$ for B equals to _____. [CUET 2022]
- (1) $\frac{\pi}{3}$ (2) $\frac{\pi}{6}$ (3) π (4) $\frac{3\pi}{2}$
02. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: The system of equations $x + y + z = 4$, $2x - y + 2z = 5$, $x - 2y - z = -3$ has unique solution.
Reason R: If A is 3×3 matrix and B is a 3×1 non-zero column matrix, then the equation $AX = B$ has unique solution if A is non-singular. [CUET 2022]
 In the light of the above statements, choose the most appropriate answer from the options given below:
 (1) Both A and R are correct and R is the correct explanation of A
 (2) Both A and R are correct and R is not the correct explanation of A
 (3) A is correct but R is not correct.
 (4) A is correct but R is correct.
03. A. If A and B are two invertible matrices, then $(AB)^{-1} = A^{-1}B^{-1}$ [CUET 2023]
 B. Every skew symmetric matrix of odd order is invertible
 C. If A is non-singular matrix, then $(A^T)^{-1} = (A^{-1})^T$
 D. If A is an involutory matrix, then $(I + A)(I - A) = 0$
 E. A diagonal matrix is both an upper triangular and a lower triangular
 Choose the correct answer from the options given below:
 (1) A, B, C, E only (2) B, D, E only (3) A, D, E only (4) C, D, E only
04. Which of the following statements are TRUE? [CUET 2024]
 (A) If each element in a row is a constant multiplier of corresponding element of another row of a determinant, then the value of the determinant is always non-zero.
 (B) If each element on one side of the principal diagonal of a determinant is zero, then the value of the determinant is the product of the diagonal elements.
 (C) The value of determinant of skew symmetric matrix of odd order is always non-zero.
 (D) If A is non-singular matrix of order three, then $|\text{adj} A| = |A|^2$
 Choose the correct answer from the options given below:
 (1) (B) and (D) only (2) (A) and (B) only
 (3) (A), (B) and (C) only (4) (A), (C) and (D) only
05. Which of the following statements are NOT TRUE? [CUET 2024]
 (A) If A and B are symmetric matrices, then $AB - BA$ is a skew symmetric matrix.
 (B) Multiplying a determinant by k means multiply elements of one column by k.
 (C) If $A^2 - A + I = 0$, then A^{-1} is equal to $A + I$.
 (D) If A and B are invertible matrices of same order, then $(A + B)^{-1} = B^{-1} + A^{-1}$.
 Choose the correct answer from the options given below:
 (1) (A), (B) and (D) only (2) (B), (C) and (D) only
 (3) (C) and (D) only (4) (A) and (C) only



CUET ALGEBRA

COMPLETE ANALYSIS



06. Math List - I with List - II

[CUET 2025]

List – I

List – II

(A) If $\begin{vmatrix} \lambda-1 & 0 \\ 0 & \lambda-1 \end{vmatrix}$, then λ is

(i) 0

(B) If $\Delta = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$, then Δ is

(ii) 1

(C) If $A = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$, then $|A^{-1}|$ is

(iii) -2

(D) If $\begin{bmatrix} a+1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$, then a is

(iv) 2

(1) (A) – (ii), (B) – (i), (C) – (iv), (D) – (iii)

(2) (A) – (ii), (B) – (i), (C) – (iii), (D) – (iv)

(3) (A) – (ii), (B) – (iii), (C) – (i), (D) – (iv)

(4) (A) – (ii), (B) – (iv), (C) – (i), (D) – (iii)



CUET ALGEBRA

COMPLETE ANALYSIS



Determinant [02]

01. If, x, y, z are all distinct and $\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$, then the value of xyz is [CUET 2023]
- (1) -2 (2) -1 (3) -3 (4) 0
02. Consider the system of linear equations as $2x + 2y + z = 1$, $4x + ky + 2z = 2$ and $kx + 4y + z = 1$ then choose the correct statement(s) from below [CUET 2024]
- (A) The system of equation has a unique solution if $k \neq 4$ and $k \neq 2$
(B) The system of equations is inconsistent for every real number k
(C) The system of equations have infinite number of solutions if $k = 4$
(D) The system of equations have infinite number of solutions if $k = 2$
- Choose the correct answer from the options given below.
- (1) (A), (B) and (D) only (2) (A), (B) and (C) only
(3) (A), (C) and (D) only (4) (C) and (D) only.
03. If $\begin{vmatrix} 1 & 1 & 1 \\ bc & ca & ab \\ a(b+c) & b(c+a) & c(a+b) \end{vmatrix} = k$, then the value of k is [CUET 2025]
- (1) $ab + bc + ca$ (2) $(ab + bc + ca)^2$ (3) $2(ab + bc + ca)$ (4) 0



CUET ALGEBRA

COMPLETE ANALYSIS



Logarithms [02]

01. Match List I with List II

[CUET 2022]

List I

List II

(a) $\log_4 (\log_3 81) =$

I. 0

(b) $3^{4\log_9 7} = 7^k$, then $k =$

II. 3

(c) $2^{\log_3 5} - 5^{\log_3 2} =$

III. 1

(d) $\log_2 [\log_2 (256)] =$

IV. 2

Choose the correct answer from the options given below:

(1) a – I; b – III; c – II; d – IV

(2) a – I; b – III; c – IV; d – II

(3) a – III; b – IV; c – II; d – I

(4) a – III; b – IV; c – I; d – II

02. The value of $e^{\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \log_{10} \tan 3^\circ + \dots + \log_{10} \tan 89^\circ}$ is

[CUET 2023]

(1) 0

(2) e

(3) 1/e

(4) 1



CUET ALGEBRA

COMPLETE ANALYSIS



Permutation & Combination [10]

01. The number of 7-digit numbers whose sum of the digits equals to 10 and which is formed by using the digit 1, 2 and 3 only is [CUET 2022]
 (1) 55 (2) 66 (3) 77 (4) 76
02. 4 Indians, 3 Americans and 2 Britishers are to be arranged around a round table. [CUET 2022]
 Answer the following questions.
 The number of ways arranging them so that the two Britishers should never come together is:
 (1) $7! \times 2!$ (2) $6! \times 2!$ (3) $7!$ (4) $6! {}^7P_2$
03. 4 Indians, 3 Americans and 2 Britishers are to be arranged around a round table. [CUET 2022]
 Answer the following questions.
 The number of ways of arranging them so that the three Americans should sit together is:
 (1) $7! \times 3!$ (2) $6! \times 3!$ (3) $6! {}^6P_3$ (4) $6! {}^7P_3$
04. 4 Indians, 3 Americans and 2 Britishers are to be arranged around a round table. [CUET 2022]
 Answer the following Questions.
 The number of ways of arranging them is
 (1) $9!$ (2) $9!/2$ (3) $8!$ (4) $8!/2$
05. Given below are two statements : [CUET 2023]
Statement I : The number of different number each of 6 digits that can be formed by using all the digits 1, 2, 1, 0, 2, 2 is 50.
Statement II : These are 4536 possibilities of writing the four digit numbers which have all distinct digits.
 In the light of the above statements, choose the correct answer from the options given below :
 (1) Both Statement I and Statement II are true (2) Both Statement I and Statement II are false
 (3) Statement I is true but Statement II is false (4) Statement I is false but Statement II is true
06. Match List I with List II [CUET 2023]
LIST I **LIST II**
 A. No. of triangles formed using 5 points in a line and 3 points on parallel line is I. 20
 B. No. of diagonals drawn using the vertices of an octagon II. 10
 C. The no. of diagonals in a regular polygon of 100 sides is III. 45
 D. A polygon with 35 diagonals has sides IV. 4850
 Choose the correct answer from the options given below:
 (1) A-I, B-II, C-III, D-IV (2) A-II, B-III, C-I, D-IV
 (3) A-III, B-IV, C-I, D-II (4) A-III, B-I, C-IV, D-II
07. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R. [CUET 2023]
Assertion A : The number of parallelograms in a chess board is 1296.
Reason R : The number of parallelograms when a set of m parallel lines is intersected by another set of n parallel lines is ${}^mC_2 \cdot {}^nC_2$.
 In the light of the above statements, choose the correct answer from the options given below :
 (1) Both A and R are true and R is the correct explanation of A
 (2) Both A and R are true but R is not the correct explanation of A
 (3) A is true but R is false
 (4) A is false but R is true

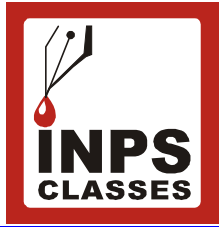


CUET ALGEBRA

COMPLETE ANALYSIS

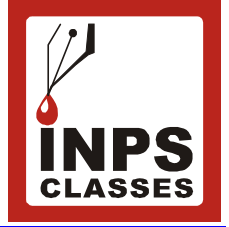


08. A person goes in for an examination in which there are four papers with a maximum of m marks from each paper. The number of ways in which one can get $2m$ marks is [CUET 2023]
- (1) $\frac{1}{3}(m+1)(2m^2 + 4m + 1)$ (2) $\frac{1}{3}(m+1)(2m^2 + 4m + 2)$
- (3) $\frac{1}{3}(m+1)(2m^2 + 4m + 3)$ (4) $^{2m+3}C_3$
09. There are 15 points in a plane such that 5 points are collinear and no three of the remaining points are collinear then total number of straight lines formed are: [CUET 2024]
- (1) 105 (2) 95 (3) 96 (4) 106
10. If permutation of the letters of the word 'AGAIN' are arranged in the order as in a dictionary then 49th word is [CUET 2024]
- (1) INGAA (2) INAAG (3) NAAGI (4) GNAAI
11. In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. If she/he attempts all 60 questions and secures 130 marks, the number of questions she/he attempts wrongly, are? [CUET 2025]
- (1) 38 (2) 22 (3) 21 (4) 37
12. Out of 5 consonants and 4 vowels, how many words of 3 consonants and 3 vowels can be made? [CUET 2025]
- (1) 40 (2) 80 (3) 20 (4) 240



CUET ALGEBRA

COMPLETE ANALYSIS



Binomial Theorem [1]

01. If $\frac{1}{9!} + \frac{1}{10!} = \frac{x}{11!}$, then the value of x is:

[CUET 2025]

(1) 121

(2) 120

(3) 12

(4) 24

INPS